Proshenjit Sarker

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Education

2024 – 2025 **M.Sc. in Electronics and Communication Engineering, Khulna University**.
Thesis title: *Integrated Surveillance System for Khulna University: Enhancing Safety, Security, and Sustainability through Camera Footage and Peripheral Sensors*.
Thesis Defense Date: 11 December 2025 (Completed)
Expected CGPA: 3.86/4.00, Total Credit: 40

2019–2023 **B.Sc. in Electronics and Communication Engineering, Khulna University**.
Four Year Honours Degree, CGPA: 3.59/4.00 (75.80% marks), Total Credits: 160.
Thesis title: *Multi-Platform Data Security Systems using n-Bit Modified (n-BM) and Reusable Block (n-RB) AES*.

Research Publications

Journal Articles

1. P. Sarker, A. Ksibi, M. M. Jamjoom, K. Choi, A. A. Nahid, and M. A. Samad, "Breast cancer prediction with feature-selected xgb classifier, optimized by metaheuristic algorithms," *Journal of Big Data*, vol. 12, no. 1, p. 78, 2025.
2. P. Sarker, K. Choi, A.-A. Nahid, and M. A. Samad, "Catboost with physics-based metaheuristics for thyroid cancer recurrence prediction," *BioData Mining*, vol. 18, no. 1, p. 84, 2025.
3. P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Metaheuristic-driven feature selection for human activity recognition on ku-har dataset using xgboost classifier," *Sensors*, vol. 25, no. 17, p. 5303, 2025.
4. P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Gallstone classification using random forest optimized by sand cat swarm optimization algorithm with shap and dice-based interpretability," *Sensors*, vol. 25, no. 17, p. 5489, 2025.
5. P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Dengue fever detection using swarm intelligence and xgboost classifier: An interpretable approach with shap and dice," *Information*, vol. 16, no. 9, p. 789, 2025.
6. P. Das, P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Breast cancer prediction using rotation forest algorithm along with finding the influential causes," *Bioengineering*, vol. 12, no. 10, p. 1020, 2025.
7. P. Das, P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Dengue fever classification integrating bird swarm algorithm with gradient boosting classifier along with feature selection and shap–dice based interpretability," *Applied Sciences*, vol. 15, no. 21, p. 11 413, 2025.
8. K. Shrestha, P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Metaheuristic-based gallstone classification using rotational forest explained with shap," *Frontiers in Digital Health*, vol. 7, p. 1 727 559,

Conference Proceedings

1. M. I. H. Sikder, P. Sarker, M. T. Hasan, and M. I. Kadir, "Reusable block aes: A modified aes algorithm for the encryption and decryption of sensitive data," in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, IEEE, 2025, pp. 1–6.

2 M. I. H. Sikder, P. Sarker, and M. I. Kadir, "A smartphone application for multiformat data security using n-bit reusable block aes," in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, IEEE, 2025, pp. 1–6.

Research Under Progress

1. Feature Reduction Using Swarm Optimization and Random Forest Classifiers for Early Diabetes Risk Prediction, *Scientific Reports* (**Accepted**)
2. Gradient Boosting Classifier with Zebra Optimization Algorithm for Pregnancy Risk Prediction, *ICT Express* (**Under Review**)
3. Swarm-Based Metaheuristic Algorithms and Classifiers for Predicting Myocardial Infarction, *The Journal of Supercomputing* (**Under Review**)
4. Swarm-Optimized Machine Learning for Rapid and Explainable Diagnosis of Rheumatic and Autoimmune Diseases, *Artificial Intelligence In Medicine* (**With Editor**)

Projects

App for Data Encryption	Developed an APK to securely encrypt images and text; part of my undergraduate thesis.
Height Measurement Machine	Arduino UNO–based non-contact height measurement using ultrasonic sensor and Android app.
CRC Simulator	Simulator for binary data transmission; indicates errors via syndromes.

Skills

Languages	Strong reading, writing and speaking competencies for English and Bangla.
Coding	Python (Classification; Metaheuristic Algorithms: SLO, WOA, FOX, BESA, EOA, EVOA, HBA, SCSO, and others; XAI: SHAP and LIME, and DiCE), Matlab, C/C++, JAVA, LaTeX.
Software	MS Office (Word, Excel, PowerPoint, Visio); basic Android Studio and Cisco Packet Tracer.
Embedded Systems	Arduino, NVIDIA Jetson Developer Kit.
Others	Academic research, journal writing, and LaTeX typesetting using Overleaf and TeXstudio, with several publications.

Experience

Research Assistant

2025 Developing Security System using Jetson Developer Kit, YOLO, Viola Jones, and ViT Algorithms.

Industrial Training

2023 Khulna 225 MW Combined Cycle Power Plant.

Certification/Award

2024 **Certificate of Appreciation**. IEEE SPICSCON 2024, for outstanding contributions as Vice Chairperson (Technical), IEEE KU SB.

Experience (continued)

2020 **Certification**. Technical Support Fundamentals, an online non-credit course authorized by Google and offered through Coursera.

2019 **Champion**. Fresher Debaters' League 2019, Khulna University.

References

Reference 01

Abdullah-Al Nahid
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Electronics and Communication Engineering Discipline, Khulna University,
Khulna, Bangladesh.
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Reference 02

Md Abdus Samad
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South Korea.
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